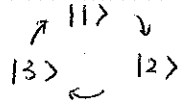


Homework 2 (1.6.1 ~ 1.8.10)

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1.6.1

$$U|1\rangle = |2\rangle, \quad U|2\rangle = |3\rangle, \quad U|3\rangle = |1\rangle$$



This is a cyclic permutation of the three vectors.

It is also equivalent to rotation of the coordinate axis along $(1,1,1)$ by 120° .

1.6.2

(1) $(\Omega\Lambda)^+ = \Lambda^+\Omega^+ = \Lambda\Omega \neq \Omega\Lambda$ not Hermitian

(2) $(\Omega\Lambda + \Lambda\Omega)^+ = (\Omega\Lambda)^+ + (\Lambda\Omega)^+ = \Lambda^+\Omega^+ + \Omega^+\Lambda^+ = \Lambda\Omega + \Omega\Lambda = \Omega\Lambda + \Lambda\Omega$
Hermitian

(3) $[\Omega, \Lambda]^+ = (\Omega\Lambda - \Lambda\Omega)^+ = (\Omega\Lambda)^+ - (\Lambda\Omega)^+ = \Lambda^+\Omega^+ - \Omega^+\Lambda^+ = \Lambda\Omega - \Omega\Lambda$
 $= -[\Omega, \Lambda]$ anti-Hermitian

(4) $(i[\Omega, \Lambda])^+ = -i[\Omega, \Lambda]^+ = -i \cdot \{-[\Omega, \Lambda]\} = i[\Omega, \Lambda]$ Hermitian

1.6.3

Suppose U_1, U_2 are unitary. $U_1^+U_1 = U_2^+U_2 = I$.

Therefore, $(U_1U_2)^+(U_1U_2) = U_2^+U_1^+U_1U_2 = U_2^+(U_1^+U_1)U_2 = U_2^+IU_2$
 $= U_2^+U_2 = I$.

Hence, U_1U_2 is unitary.

1.6.4

$$\det(U^+U) = \det I = 1$$

$$\det(U^+U) = \det(U^+)\det U = \det(U^T)^*\det U = (\det U)^*(\det U)$$

$$= |\det U|^2$$

Therefore, $|\det U| = 1$, which indicates that $\det U$ is a complex number of unit modulus.

1.6.5

According to Example 1.6.1, $R(\frac{1}{2}\pi i) = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}$.

$$\text{Therefore, } R^+ R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I.$$

Hence, $R(\frac{1}{2}\pi i)$ is unitary.

Unitary

1.6.6

$$\textcircled{1} \quad \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}^+ \cdot \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = I$$

$$\det \left[\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \right] = \left(\frac{1}{\sqrt{2}} \right)^2 (1 \cdot 1 - i \cdot i) = 1 = e^{i\theta}, \text{ when } \theta = 2k\pi \text{ for integer } k.$$

$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix}^+ = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & -i \\ -i & 1 \end{pmatrix} \neq \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & i \\ i & 1 \end{pmatrix} \quad \text{Not Hermitian.}$$

$$\textcircled{2} \quad \frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}^+ \cdot \frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix} = \frac{1}{4} \begin{pmatrix} 1-i & 1+i \\ 1+i & 1-i \end{pmatrix} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}$$

$$= \frac{1}{4} \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix} = I \quad \text{Unitary}$$

$$\det \left[\frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix} \right] = \left(\frac{1}{2} \right)^2 [(1+i)^2 - (1-i)^2] = \frac{1}{4} (2i + 2i) = i$$

$$= e^{i\theta}, \text{ when } \theta = 2k\pi + \frac{\pi}{2} \text{ for integer } k.$$

$$\frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix}^+ = \frac{1}{2} \begin{pmatrix} 1-i & 1+i \\ 1+i & 1-i \end{pmatrix} \neq \frac{1}{2} \begin{pmatrix} 1+i & 1-i \\ 1-i & 1+i \end{pmatrix} \quad \text{Not Hermitian}$$

1.7.1

$$(1) \quad \text{Tr}(\Omega \Lambda) = \sum_i (\Omega \Lambda)_{ii} = \sum_i \sum_j \Omega_{ij} \Lambda_{ji} = \sum_j \sum_i \Lambda_{ji} \Omega_{ij} = \sum_j (\Lambda \Omega)_{jj} = \text{Tr}(\Lambda \Omega).$$

$$(2) \quad \text{Tr}(\Omega \Lambda \Theta) = \sum_i (\Omega \Lambda \Theta)_{ii} = \sum_i \sum_j \sum_k \Omega_{ij} \Lambda_{jk} \Theta_{ki} = \sum_j \sum_k \sum_i \Lambda_{jk} \Theta_{ki} \Omega_{ij}$$

$$= \sum_j (\Lambda \Theta \Omega)_{jj} = \text{Tr}(\Lambda \Theta \Omega)$$

$$= \sum_k \sum_i \sum_j \Theta_{ki} \Omega_{ij} \Lambda_{jk} = \sum_k (\Theta \Omega \Lambda)_{kk} = \text{Tr}(\Theta \Omega \Lambda)$$

$$(3) \quad \text{Tr}(U^+ \Omega U) = \text{Tr}(\Omega U U^+) = \text{Tr}(\Omega I) = \text{Tr} \Omega.$$

1.7.2

$$\begin{aligned}
 \det(U^\dagger \Omega U) &= \det U^\dagger \det \Omega \det U \\
 &= \det \Omega (\det U^\dagger \det U) \\
 &= \det \Omega \det(U^\dagger U) \\
 &= \det \Omega \cdot 1 \\
 &= \det \Omega.
 \end{aligned}$$

1.8.1

$$(1) \det(\Omega - I\omega) = \begin{vmatrix} 1-\omega & 3 & 1 \\ 0 & 2-\omega & 0 \\ 0 & 1 & 4-\omega \end{vmatrix} = (1-\omega)(2-\omega)(4-\omega) = 0$$

$$\omega = 1, 2, 4$$

$$\omega = 1: \begin{pmatrix} 0 & 3 & 1 \\ 0 & 1 & 0 \\ 0 & 1 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \Rightarrow |\omega=1\rangle = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\omega = 2: \begin{pmatrix} -1 & 3 & 1 \\ 0 & 0 & 0 \\ 0 & 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \Rightarrow |\omega=2\rangle = \frac{1}{\sqrt{30}} \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}$$

$$\omega = 4: \begin{pmatrix} -3 & 3 & 1 \\ 0 & -2 & 0 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = 0 \Rightarrow |\omega=4\rangle = \frac{1}{\sqrt{10}} \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$$

$$(2) \Omega^\dagger \neq \Omega \quad \text{NOT Hermitian}$$

$$\langle \omega=1 | \omega=2 \rangle \neq 0 \quad \langle \omega=1 | \omega=4 \rangle \neq 0 \quad \langle \omega=2 | \omega=4 \rangle \neq 0.$$

The eigenvectors are not orthogonal.

1.8.2

$$(1) \Omega^\dagger = \Omega \quad \text{Hermitian}$$

$$(2) \det(\Omega - I\omega) = \begin{vmatrix} -\omega & 0 & 1 \\ 0 & -\omega & 0 \\ 1 & 0 & -\omega \end{vmatrix} = -\omega(\omega^2 - 1) = 0$$

$$\omega = 0, 1, -1.$$

$$\omega = 0 : \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Rightarrow |\omega=0\rangle = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\omega = 1 : \begin{pmatrix} -1 & 0 & 1 \\ 0 & -1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Rightarrow |\omega=1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$\omega = -1 : \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Rightarrow |\omega=-1\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

$$(3) \quad U = \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \quad U^\dagger = \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

$$U^\dagger \Omega U = \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

1.8.3

$$(1) \quad \det(\Omega - I\omega) = \begin{vmatrix} 1-\omega & 0 & 0 \\ 0 & \frac{3}{2}-\omega & -\frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{3}{2}-\omega \end{vmatrix} = (1-\omega) \left[\left(\frac{3}{2}-\omega \right)^2 - \frac{1}{4} \right] = 0$$

$$\omega = 1 \quad \text{or} \quad \omega^2 - 3\omega + 2 = 0$$

$$\Rightarrow \omega_1 = \omega_2 = 1, \quad \omega_3 = 2.$$

$$(2) \quad \omega = 2 : \begin{pmatrix} -1 & 0 & 0 \\ 0 & -\frac{1}{2} & -\frac{1}{2} \\ 0 & -\frac{1}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Rightarrow \begin{aligned} x &= 0 \\ y+z &= 0 \end{aligned}$$

Set $y = a$, we have $z = -a$.

$$\text{Therefore, } |\omega=2\rangle = \frac{1}{\sqrt{2}a^2} \begin{pmatrix} 0 \\ a \\ -a \end{pmatrix}$$

$$(3) \quad \omega = 1 : \begin{pmatrix} 0 & 0 & 0 \\ 0 & \frac{1}{2} & -\frac{1}{2} \\ 0 & -\frac{1}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Rightarrow \begin{aligned} x &\text{ is arbitrary, set it to be } b, \\ y-z &= 0, \text{ Set } y=c, \text{ we have } z=c. \end{aligned}$$

$$\text{Therefore, } |\omega=1\rangle = \frac{1}{\sqrt{b^2+2c^2}} \begin{pmatrix} b \\ c \\ c \end{pmatrix}.$$

1.8.4

$$(1) \det(\Omega - I\omega) = \begin{vmatrix} 4-\omega & 1 \\ -1 & 2-\omega \end{vmatrix} = \omega^2 - 6\omega + 9 = 0$$

$$\omega_1 = \omega_2 = 3$$

$$(2) \omega = 3: \begin{pmatrix} 1 & 1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \Rightarrow x + y = 0$$

Set $x = a$, we have $y = -a$.

$$\text{Therefore, } |\omega = 3\rangle = \frac{1}{\sqrt{2}a^2} \begin{pmatrix} a \\ -a \end{pmatrix}.$$

This is the only eigenvector we can find.

1.8.5

$$(1) \Omega^\dagger \Omega = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

$$= \begin{pmatrix} \cos^2\theta + \sin^2\theta & \cos\theta\sin\theta - \sin\theta\cos\theta \\ \sin\theta\cos\theta - \cos\theta\sin\theta & \sin^2\theta + \cos^2\theta \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I$$

$$(2) \det(\Omega - I\omega) = \begin{vmatrix} \cos\theta - \omega & \sin\theta \\ -\sin\theta & \cos\theta - \omega \end{vmatrix} = \omega^2 - 2\omega\cos\theta + 1 = 0$$

$$\omega = \cos\theta \pm i\sin\theta = e^{\pm i\theta}$$

$$(3) \omega = e^{i\theta}: \begin{pmatrix} \cos\theta - e^{i\theta} & \sin\theta \\ -\sin\theta & \cos\theta - e^{i\theta} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -i\sin\theta & \sin\theta \\ -\sin\theta & -i\sin\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$$\Rightarrow -ix + y = 0 \Rightarrow |\omega = e^{i\theta}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$\omega = e^{-i\theta}: \begin{pmatrix} \cos\theta - e^{-i\theta} & \sin\theta \\ -\sin\theta & \cos\theta - e^{-i\theta} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} i\sin\theta & \sin\theta \\ -\sin\theta & i\sin\theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$$\Rightarrow ix + y = 0 \Rightarrow |\omega = e^{-i\theta}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$\langle \omega = e^{i\theta} | \omega = e^{-i\theta} \rangle = \frac{1}{2} (1 - i) \begin{pmatrix} 1 \\ -i \end{pmatrix} = \frac{1}{2} (1 + i^2) = 0.$$

Orthogonal.

$$(4) \quad U = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \end{pmatrix}$$

$$U^\dagger \Omega U = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{i}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{i}{\sqrt{2}} & -\frac{i}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{-i\theta} \end{pmatrix}$$

1.8.6

(1) Suppose U is the unitary matrix that transforms Ω in to diagonal matrix.

$$\det \Omega = \det(U^\dagger \Omega U) = \det D = \prod_{i=1}^n \omega_i.$$

Where D is the corresponding diagonal matrix with Ω 's eigenvalues ω_i on its diagonal.

$$(2) \quad \text{Tr } \Omega = \text{Tr}(U^\dagger \Omega U) = \text{Tr } D = \sum_{i=1}^n \omega_i.$$

1.8.7

According to 1.8.6, we have

$$\begin{cases} \omega_1 + \omega_2 = \text{Tr } \Omega = 1 + 1 = 2 \\ \omega_1 \omega_2 = \det \Omega = 1 \times 1 - 2 \times 2 = -3 \end{cases}$$

$$\Rightarrow \begin{cases} \omega_1 = -1 \\ \omega_2 = 3 \end{cases} \quad \text{or} \quad \begin{cases} \omega_1 = 3 \\ \omega_2 = -1 \end{cases}.$$

Verification.

$$\det(\Omega - I\omega) = \begin{vmatrix} 1-\omega & 2 \\ 2 & 1-\omega \end{vmatrix} = 0$$

$$\Rightarrow \omega^2 - 2\omega - 3 = 0 \Rightarrow \omega_1 = 3, \omega_2 = -1.$$

1.8.8

$$(1) \quad M^i M^j + M^j M^i = 2\delta_{ij} I \Rightarrow M^i M^i = I$$

$$\text{Therefore, } M^i M^i |\omega\rangle = M^i (\omega |\omega\rangle) = \omega^2 |\omega\rangle$$

$$\text{and } M^i M^i |\omega\rangle = I |\omega\rangle = |\omega\rangle.$$

$$\Rightarrow \omega^2 = 1 \Rightarrow \omega = \pm 1.$$

$$(2) \quad M^i M^j = -M^j M^i$$

$$\Rightarrow M^j M^i M^j = -M^j M^j M^i = -M^i$$

$$\begin{aligned} \Rightarrow \text{Tr } M^i &= \text{Tr}(-M^j M^i M^j) = -\text{Tr}(M^j M^i M^j) \\ &= -\text{Tr}(M^i M^j M^j) = -\text{Tr}(M^i I) \\ &= -\text{Tr } M^i \end{aligned}$$

$$\Rightarrow \text{Tr } M^i = 0.$$

(3) According to 1.8.6

$$\text{Tr } M^i = \sum_{k=1}^n \omega_k, \text{ where } n \text{ is the dimension of the matrix.}$$

Since $\omega_k = \pm 1$, $\text{Tr } M^i$ can be zero only if n is even.

* The sum of an odd number of odd numbers is still odd, and cannot be zero.

1.8.9.

$$\begin{aligned} \vec{L} &= \sum_{\alpha} m_{\alpha} \vec{r}_{\alpha} \times (\vec{\omega} \times \vec{r}_{\alpha}) \\ &= \sum_{\alpha} m_{\alpha} [\vec{\omega} (\vec{r}_{\alpha} \cdot \vec{r}_{\alpha}) - \vec{r}_{\alpha} (\vec{r}_{\alpha} \cdot \vec{\omega})] \\ &= \sum_{\alpha} m_{\alpha} [\vec{\omega} \cdot r_{\alpha}^2 - \vec{r}_{\alpha} (\vec{r}_{\alpha} \cdot \vec{\omega})] \\ \Rightarrow L_i &= \sum_{\alpha} m_{\alpha} [\omega_i r_{\alpha}^2 - (\vec{r}_{\alpha})_i (\vec{r}_{\alpha} \cdot \vec{\omega})] \\ &= \sum_{\alpha} m_{\alpha} [\omega_i r_{\alpha}^2 - (\vec{r}_{\alpha})_i \sum_j (\vec{r}_{\alpha})_j \omega_j] \\ &= \sum_j \sum_{\alpha} m_{\alpha} [r_{\alpha}^2 \delta_{ij} - (\vec{r}_{\alpha})_i (\vec{r}_{\alpha})_j] \omega_j \\ &\equiv \sum_j M_{ij} \omega_j, \end{aligned}$$

$$\text{where } M_{ij} = \sum_{\alpha} m_{\alpha} [r_{\alpha}^2 \delta_{ij} - (\vec{r}_{\alpha})_i (\vec{r}_{\alpha})_j].$$

$$\text{Or } |L\rangle = M |\omega\rangle.$$

(1) No. The angular momentum and angular velocity are not parallel unless $|\omega\rangle$ is an eigenvector of M .

$$(2) \quad M_{ji}^* = \left(\sum_{\alpha} m_{\alpha} [r_{\alpha}^2 \delta_{ji} - (\vec{r}_{\alpha})_j (\vec{r}_{\alpha})_i] \right)^*$$

Since m_{α} , r_{α}^2 , $(\vec{r}_{\alpha})_j$, $(\vec{r}_{\alpha})_i$ are all real,

$$M_{ji}^* = \sum_{\alpha} m_{\alpha} [r_{\alpha}^2 \delta_{ji} - (\vec{r}_{\alpha})_j (\vec{r}_{\alpha})_i] = \sum_{\alpha} m_{\alpha} [r_{\alpha}^2 \delta_{ij} - (\vec{r}_{\alpha})_i (\vec{r}_{\alpha})_j] = M_{ij}.$$

Hermitian.

(3) Since M is Hermitian, we can always find three eigenvectors which are orthogonal to each other by solving the eigen-problem $M|\omega\rangle = \omega|\omega\rangle$. And these three eigenvectors denote the three directions for $\vec{\omega}$ we want to find in the 3-D Euclidean space.

(4) The complete symmetry of sphere means all directions are equivalent eigendirections. Therefore, the eigenvalues are degenerate.

1.8.10

$$[\Omega, \Lambda] = \Omega\Lambda - \Lambda\Omega = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 2 \end{pmatrix} - \begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 3 & 0 & 3 \\ 0 & 0 & 0 \\ 3 & 0 & 3 \end{pmatrix} - \begin{pmatrix} 3 & 0 & 3 \\ 0 & 0 & 0 \\ 3 & 0 & 3 \end{pmatrix} = 0$$

Since the two Hermitian matrices commute, they can be diagonalized simultaneously.

Starting from Λ ,

$$\det(\Lambda - I\omega) = \begin{vmatrix} 2-\omega & 1 & 1 \\ 1 & -\omega & -1 \\ 1 & -1 & 2-\omega \end{vmatrix} = (2-\omega)(\omega+1)(\omega-3) = 0$$

$$\omega = -1, 2, 3.$$

$$\omega = -1: \begin{pmatrix} 3 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Rightarrow |\omega = -1\rangle = \frac{1}{\sqrt{6}} \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$$

$$\omega = 2: \begin{pmatrix} 0 & 1 & 1 \\ 1 & -2 & -1 \\ 1 & -1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Rightarrow |\omega = 2\rangle = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$$

$$\omega = 3: \begin{pmatrix} -1 & 1 & 1 \\ 1 & -3 & -1 \\ 1 & -1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 0 \Rightarrow |\omega = 3\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

$$U = \begin{pmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \\ -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 \\ -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \end{pmatrix}$$

$$U^\dagger \Lambda U = \begin{pmatrix} \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 2 & 1 & 1 \\ 1 & 0 & -1 \\ 1 & -1 & 2 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \\ -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 \\ -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{pmatrix},$$

$$U^\dagger \Omega U = \begin{pmatrix} \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \\ -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 \\ -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$